Computer Science Project Contract

Leiden Institute of Advanced Computer Science
Faculty of Science, Leiden University

Agreement on Research Project/Master Thesis between:

Student

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Supervisor(s)

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Research project

Project summary:	In this thesis, we will be analyzing Reinforcement Learning (RL) baselines frameworks. These are repositories with a collection of RL algorithms that serve as a purpose to be consistent, useful for easy reproduction and convenience for testing, and reliability for comparison against other algorithms. However, due to rapid growth of the collection of algorithms and number of baselines frameworks available, it is at question whether the RL algorithms provided perform similarly. This thesis includes analysis of various baselines frameworks and algorithms on a fixed set of environments to compare whether the performance matches the performance of the original paper the algorithm was published, whether the results obtained from these baselines frameworks may be directly comparable to each other, and whether the relative ranking of RL algorithms within a baseline framework is preserved across other baselines frameworks.
Working title:	How Stable are Baselines**:** A study on the reliability of Reinforcement Learning baselines
Date of project start:	19-02-2025
Deadline for handing in the report:	31-08-2025

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Initial project plan / global planning:

High level work breakdown:	First, we do literary research on this problem. This includes retrieving what baseline frameworks are currently being used, what RL algorithms are implemented within these baselines, what key differences there are between baseline frameworks, what order of performance is reported when addressing algorithms within a baseline framework, what hyperparameter settings are used within an algorithm and whether that is comparable to the original paper where this algorithm was introduced, what environments are commonly used for reporting performance of an algorithm, and whether the repositories of the environments are concurrent. The more information gathered on this matter, the more of an overview we retrieve on the magnitude of the problem. Furthermore, we will retrieve perform repeated experiments using different baselines frameworks and algorithms on a fixed set of environments (assumingly gathered via the literary research). We will use hyperparameters reported in the original paper (or if unavailable, the best hyperparameter settings from the baseline frameworks itself, applied on all algorithms). Via this, we can establish whether the performance of algorithms between baseline frameworks is similar, and whether the ranking of algorithms between baseline frameworks is the same.

The global planning is as follows:

Feb-Apr 2025	I will perform literary research, as described in the high level work breakdown. In this time, I will also familiarize with the algorithms used within this work, and set up corresponding pipelines. At best, a first step of experimenting can be done. This includes using one algorithm (likely to be PPO), from n baselines, and test it on m environments.
Apr-Jun 2025	I will continue to work on experimenting, using other algorithms, to establish the similarity of performance of multiple algorithms between baseline frameworks and relative ranking of algorithms between baseline frameworks.
Jun-Aug 2025	Depending on the state of the results and our pending hypothesis, either to dive deeper in experimenting. This time period is also used to write the thesis.

Other agreements:

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Signatures

Date: 19-02-2025

Location: Leiden

Student	**Supervisor**
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